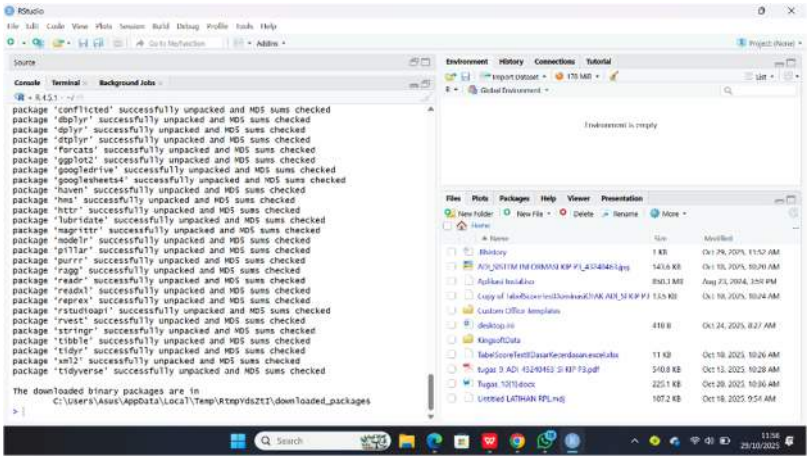
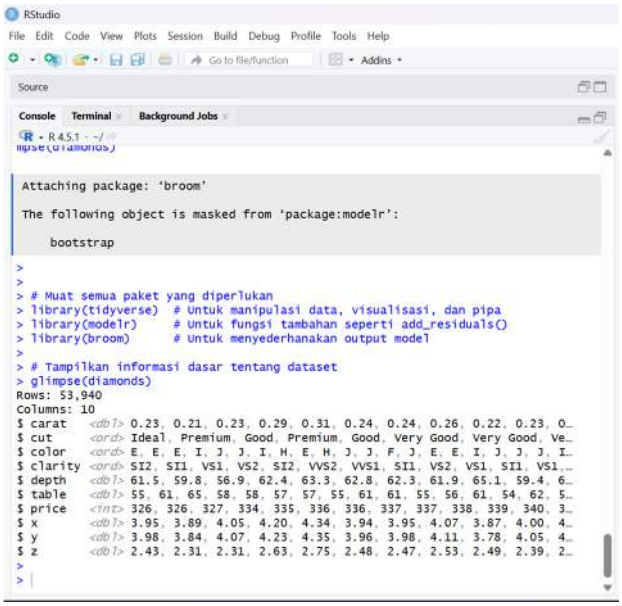
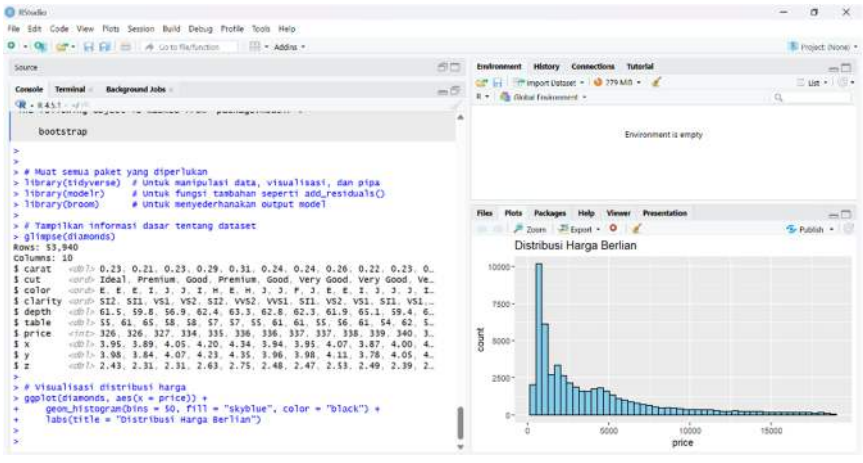
NAMA : NOVA FITRIYANTI

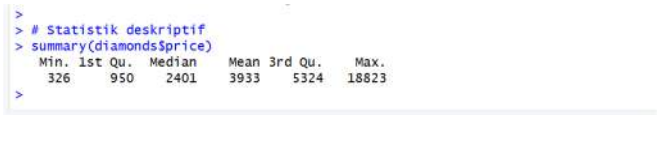
NIM : 43240417

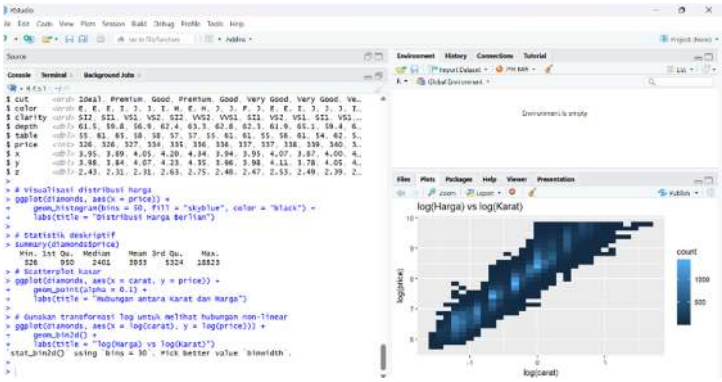
KELAS : SISTEM INFORMASI-KIP-P3











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R 4.5.1 · ~/

```
+     filter(carat <= 2.5)  # Hanya 99.7% data terbesar
> diamonds2 <- diamonds2 %>%
+     mutate(lprice = log2(price),
+            lcarat = log2(carat))
>
> diamonds2 <- diamonds %>%
+     filter(carat <= 2.5)  # Hanya 99.7% data terbesar
> diamonds2 <- diamonds2 %>%
+     mutate(lprice = log2(price),
+            lcarat = log2(carat))
> # Periksa nilai x, y, z yang tidak masuk akal
> diamonds2 %>%
+     filter(y < 3 | y > 20 | x < 3 | z < 3) %>%
+     select(carat, x, y, z, price)
# A tibble: 16,585 × 5
   carat     x     y     z price
   <dbl> <dbl> <dbl> <dbl> <int>
 1  0.23  3.95  3.98  2.43   326
 2  0.21  3.89  3.84  2.31   326
 3  0.23  4.05  4.07  2.31   327
 4  0.29  4.2   4.23  2.63   334
 5  0.31  4.34  4.35  2.75   335
 6  0.24  3.94  3.96  2.48   336
 7  0.24  3.95  3.98  2.47   336
 8  0.26  4.07  4.11  2.53   337
 9  0.22  3.87  3.78  2.49   337
10  0.23  4     4.05  2.39   338
# ℹ 16,575 more rows
# ℹ Use `print(n = ...)` to see more rows
>
>
```

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R 4.5.1 · ~/

```
> mod2 <- lm(lprice ~ lcarat + cut, data = diamonds2)
>
> # Model 3: tambahkan color dan clarity
> mod3 <- lm(lprice ~ lcarat + cut + color + clarity, data = diamonds2)
>
> # Lihat koefisien dari mod3
> tidy(mod3) %>%
+     arrange(desc(abs(estimate))) %>%  # Urutkan berdasarkan besar efek
+     print(n = 15)
# A tibble: 19 × 5
   term        estimate std.error statistic   p.value
   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
 1 (Intercept)  12.2      0.00169   7212.    0
 2 lcarat        1.89     0.00112   1678.    0
 3 clarity.L     1.31     0.00518    253.    0
 4 color.L      -0.634    0.00291   -218.    0
 5 clarity.Q    -0.334    0.00484    -69.0   0
 6 clarity.C     0.178    0.00414     43.1   0
 7 cut.L         0.174    0.00339     51.3   0
 8 color.Q      -0.138    0.00268    -51.4   0
 9 clarity^4    -0.0881   0.00330    -26.7   5.33e-156
10 cut.Q        -0.0503   0.00298    -16.9   6.89e- 64
11 clarity^7     0.0482   0.00205     23.5   1.28e-121
12 clarity^5     0.0359   0.00268     13.4   8.12e- 41
13 color.C      -0.0221   0.00250     -8.82  1.19e- 18
14 cut.C         0.0191   0.00258      7.41  1.31e- 13
15 color^4       0.0166   0.00230      7.21  5.54e- 13
# ℹ 4 more rows
# ℹ Use `print(n = ...)` to see more rows
>
>
```

RStudio

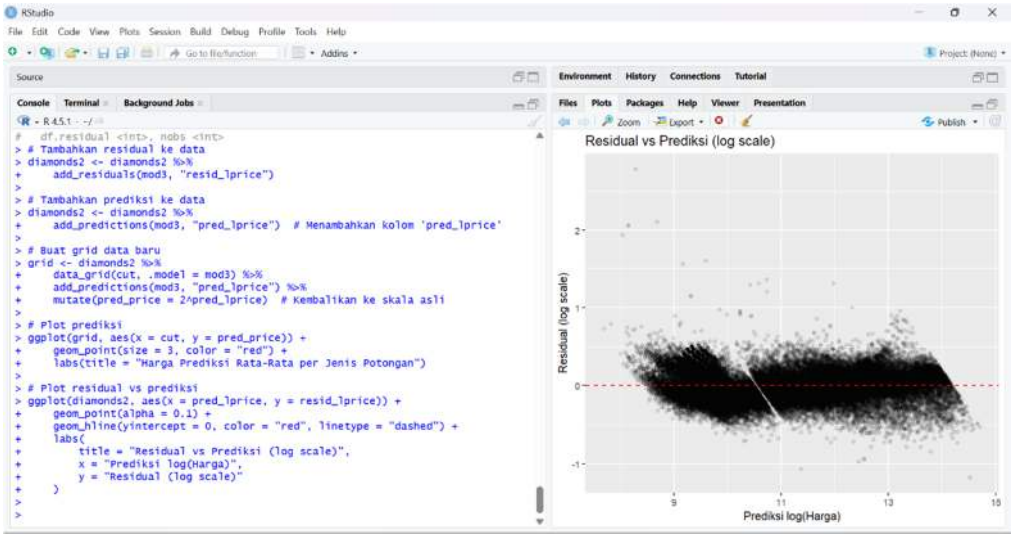
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Source

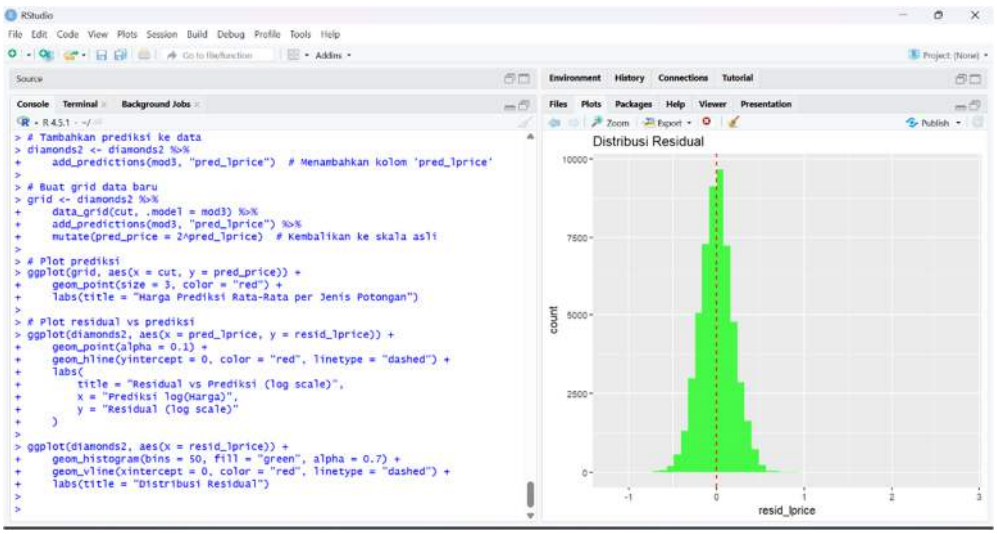
Console Terminal Background Jobs

R 4.5.1 · ~/

```
   <chr>           <dbl>     <dbl>     <dbl>      <dbl>
 1 (Intercept)  12.2      0.00169   7212.    0
 2 lcarat        1.89     0.00112   1678.    0
 3 clarity.L     1.31     0.00518    253.    0
 4 color.L      -0.634    0.00291   -218.    0
 5 clarity.Q    -0.334    0.00484    -69.0   0
 6 clarity.C     0.178    0.00414     43.1   0
 7 cut.L         0.174    0.00339     51.3   0
 8 color.Q      -0.138    0.00268    -51.4   0
 9 clarity^4    -0.0881   0.00330    -26.7   5.33e-156
10 cut.Q        -0.0503   0.00298    -16.9   6.89e- 64
11 clarity^7     0.0482   0.00205     23.5   1.28e-121
12 clarity^5     0.0359   0.00268     13.4   8.12e- 41
13 color.C      -0.0221   0.00250     -8.82  1.19e- 18
14 cut.C         0.0191   0.00258      7.41  1.31e- 13
15 color^4       0.0166   0.00230      7.21  5.54e- 13
# i 4 more rows
# i Use `print(n = ...)` to see more rows
>
> glance(mod3)
# A tibble: 1 x 12
  r.squared adj.r.squared sigma
      <dbl>         <dbl> <dbl>
1     0.983         0.983 0.192
# i 9 more variables:
#   statistic <dbl>, p.value <dbl>,
#   df <dbl>, logLik <dbl>,
#   AIC <dbl>, BIC <dbl>,
#   deviance <dbl>,
#   df.residual <int>, nobs <int>
>
```




```
#   df.residual <int>, nobs <int>
> # Tambahkan residual ke data
> diamonds2 <- diamonds2 %>%
+     add_residuals(mod3, "resid_lprice")
>
> # Tambahkan prediksi ke data
> diamonds2 <- diamonds2 %>%
+     add_predictions(mod3, "pred_lprice")  # Menambahkan kolom 'pred_lprice'
>
> # Buat grid data baru
> grid <- diamonds2 %>%
+     data_grid(cut, .model = mod3) %>%
+     add_predictions(mod3, "pred_lprice") %>%
+     mutate(pred_price = 2^pred_lprice)  # Kembalikan ke skala asli
>
> # Plot prediksi
> ggplot(grid, aes(x = cut, y = pred_price)) +
+     geom_point(size = 3, color = "red") +
+     labs(title = "Harga Prediksi Rata-Rata per Jenis Potongan")
>
> # Plot residual vs prediksi
> ggplot(diamonds2, aes(x = pred_lprice, y = resid_lprice)) +
+     geom_point(alpha = 0.1) +
+     geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
+     labs(
+         title = "Residual vs Prediksi (log scale)",
+         x = "Prediksi log(Harga)",
+         y = "Residual (log scale)"
+     )
>
>
```




```
> # Tambahkan prediksi ke data
> diamonds2 <- diamonds2 %>%
+     add_predictions(mod3, "pred_lprice")  # Menambahkan kolom 'pred_lprice'
>
> # Buat grid data baru
> grid <- diamonds2 %>%
+     data_grid(cut, .model = mod3) %>%
+     add_predictions(mod3, "pred_lprice") %>%
+     mutate(pred_price = 2^pred_lprice)  # Kembalikan ke skala asli
>
> # Plot prediksi
> ggplot(grid, aes(x = cut, y = pred_price)) +
+     geom_point(size = 3, color = "red") +
+     labs(title = "Harga Prediksi Rata-Rata per Jenis Potongan")
>
> # Plot residual vs prediksi
> ggplot(diamonds2, aes(x = pred_lprice, y = resid_lprice)) +
+     geom_point(alpha = 0.1) +
+     geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
+     labs(
+         title = "Residual vs Prediksi (log scale)",
+         x = "Prediksi log(Harga)",
+         y = "Residual (log scale)"
+     )
>
> ggplot(diamonds2, aes(x = resid_lprice)) +
+     geom_histogram(bins = 50, fill = "green", alpha = 0.7) +
+     geom_vline(xintercept = 0, color = "red", linetype = "dashed") +
+     labs(title = "Distribusi Residual")
>
>
```

```
> rsquare(mod1, diamonds2) = 0.9334026

Error in rsquare(mod1, diamonds2) = 0.9334026 :
  could not find function "rsquare<-"
```

